

When is expensive perception worth paying for?

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Same page, two ways an agent sees it.

● ● ● * Claude Code Playwright MCP

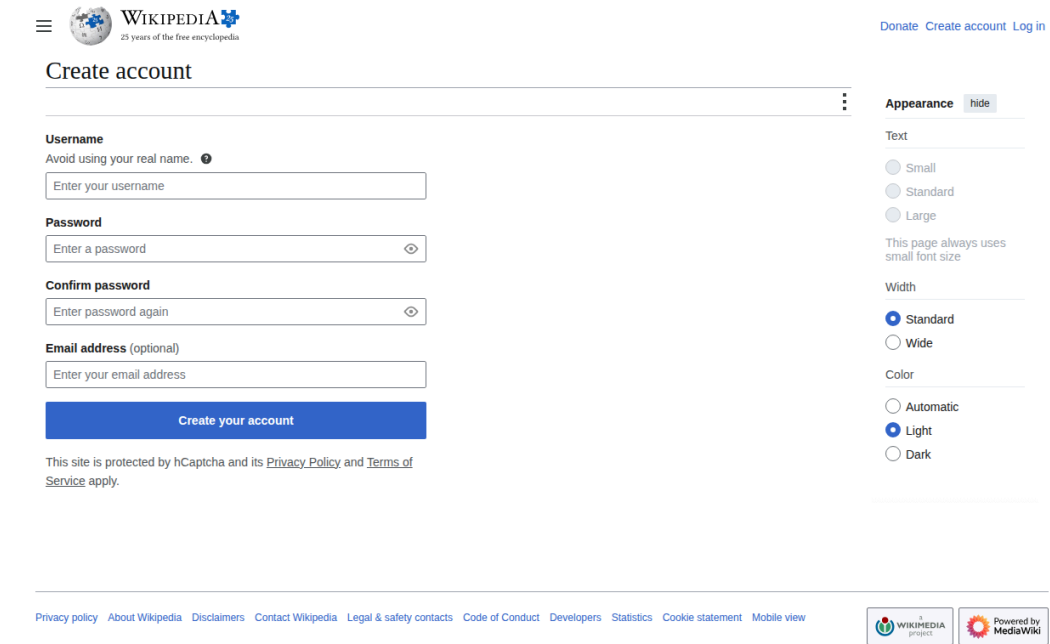
playwright · browser_snapshot - real reply, excerpt161 lines for this page

```
- generic [active] [ref=e1]:
: 26 lines
  - heading "Create account" [level=1] [ref=e35]
: 55 lines
  - generic "Create account" [ref=e115]:
    - generic [ref=e119]:
      - generic [ref=e120]:
        - generic [ref=e121]:
          - generic [ref=e122]: Username
          - generic [ref=e123]:
            - text: Avoid using your real name.
            - button "username policy" [ref=e124]
        - textbox "Username" [ref=e128]:
          - /placeholder: Enter your username
      - generic [ref=e129]:
        - generic [ref=e130]: Password
        - generic [ref=e133]:
          - textbox "Password" [ref=e134]:
            - /placeholder: Enter a password
            - generic [ref=e135] [cursor=pointer]
      - generic [ref=e136]:
        - generic [ref=e137]: Confirm password
        - generic [ref=e140]:
          - textbox "Confirm password" [ref=e141]:
            - /placeholder: Enter password again
            - generic [ref=e142] [cursor=pointer]
      - generic [ref=e143]:
        - generic [ref=e144]: Email address (optional)
        - textbox "Email address (optional)" [ref=e148]:
          - /placeholder: Enter your email address
      - button "Create your account" [ref=e151]
: 51 more lines
```

READ · the page as text

> _ Codex GPT-6 Astra · computer use

screenshot · real capture



LOOK · a screenshot

Real captures of one page, 15 Sep 2026 · frames name the tools · each can also use the other view

Holistic AI × UCL CDI Research Showcase · 16 September 2026

■ If it always picked the right view:

+11 to +16

solved per 100

−14 to −27%

token bill

−7 to −29%

CO₂e (estimated)

up to 34%

faster on one site; same on two

Perfect router vs best single view · our benchmark · large open model · not Claude or GPT results

ONE RECORDED RUN PER VIEW · NOTHING ON THIS SCREEN IS LIVE

■ Same task, three ways of seeing it, three different bills.

IF YOU BUILD A WEB AGENT

Should it look, read, or both — and can it learn to choose?

1

How do the views behave?

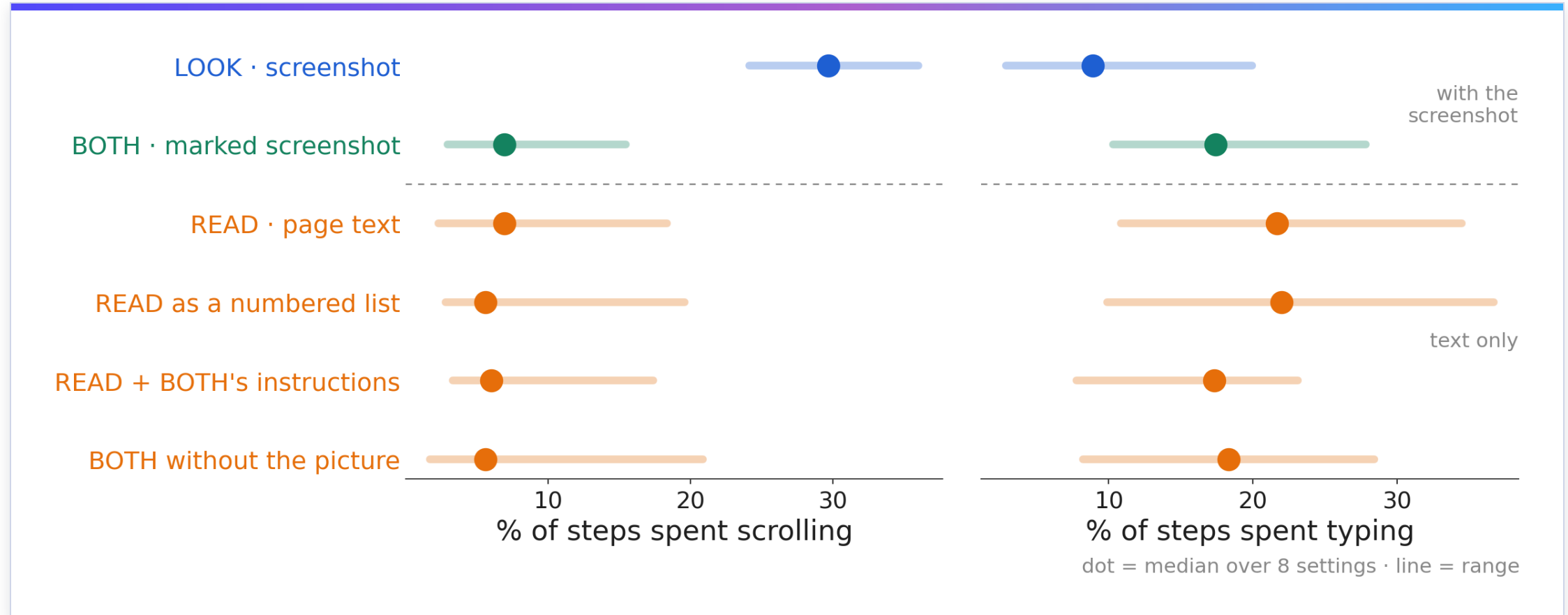
2

How do they fail?

3

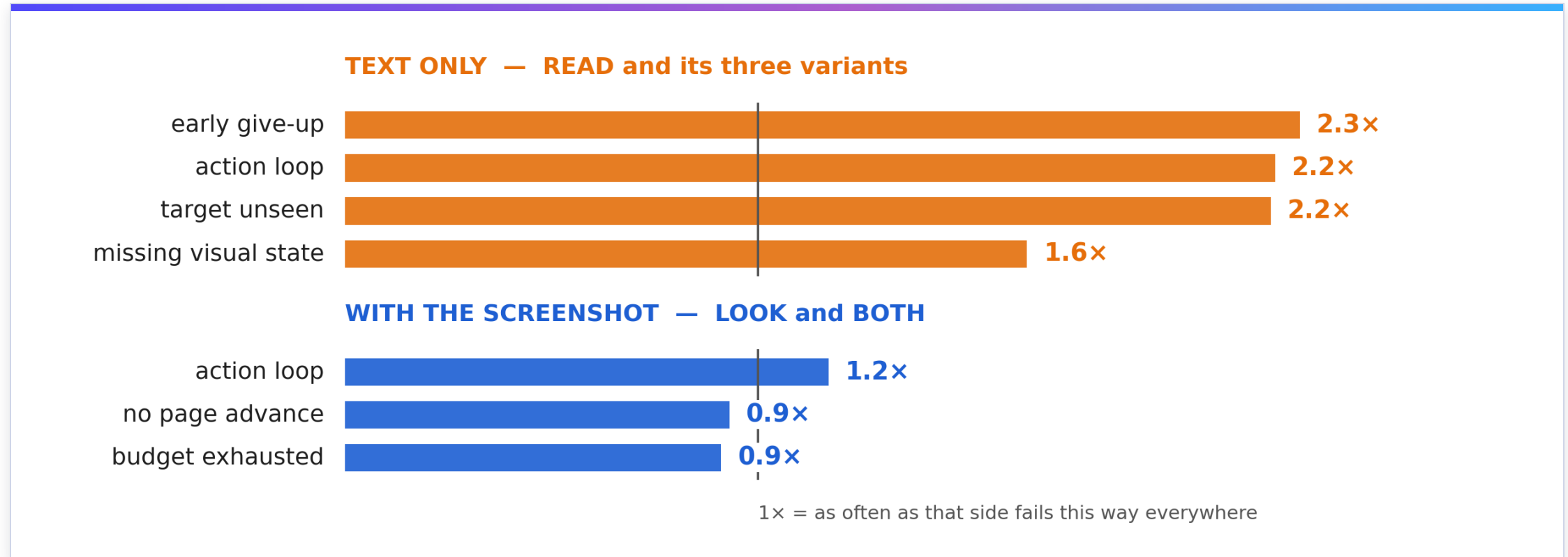
Can it be learned?

■ With only the screenshot, the agent scrolls far more and types far less.



Medians over 8 settings: LOOK scrolls in **about 30%** of its steps, every other view **about 6%**; it types about half as often

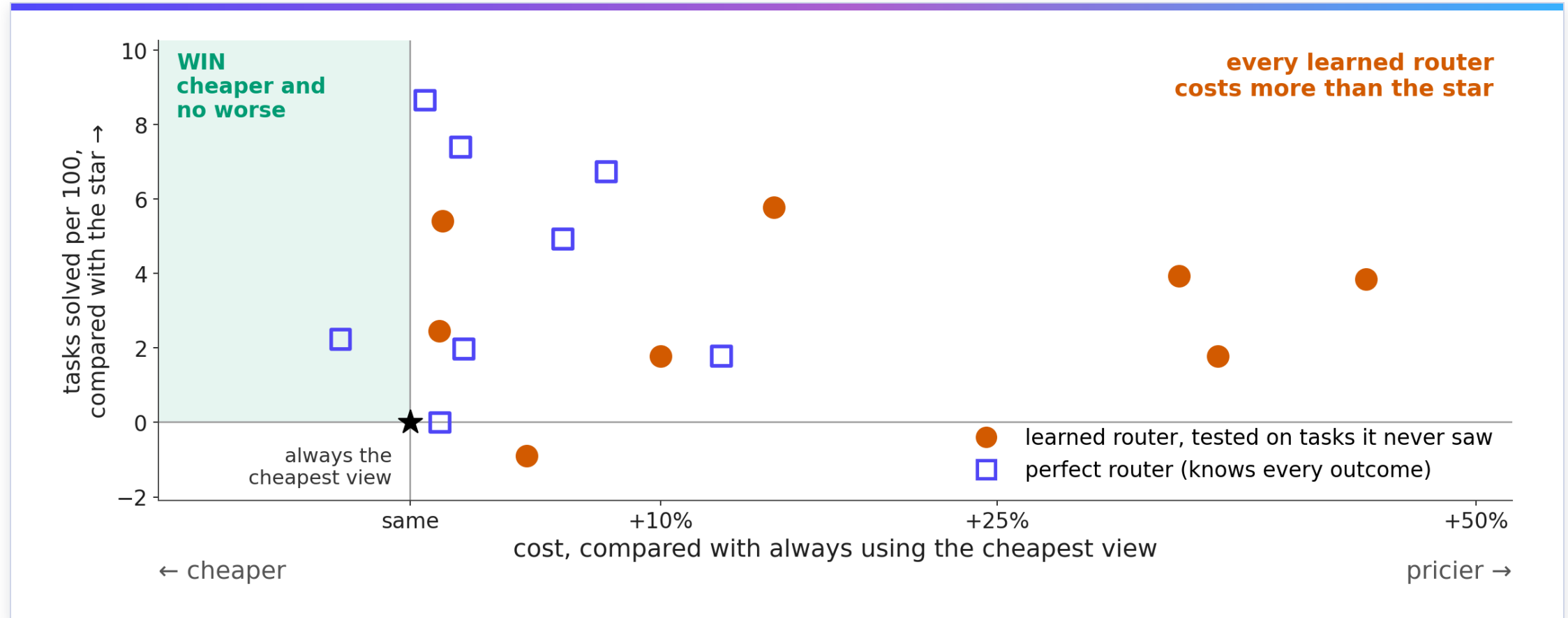
- Text-only views fail in ways you can name; views with the screenshot just never get there.



Each side vs its own failures elsewhere · text-only: several over **2x** · LOOK and BOTH: none stands out · six VisualWebArena settings

③ CAN A ROUTER LEARN WHICH VIEW TO USE?

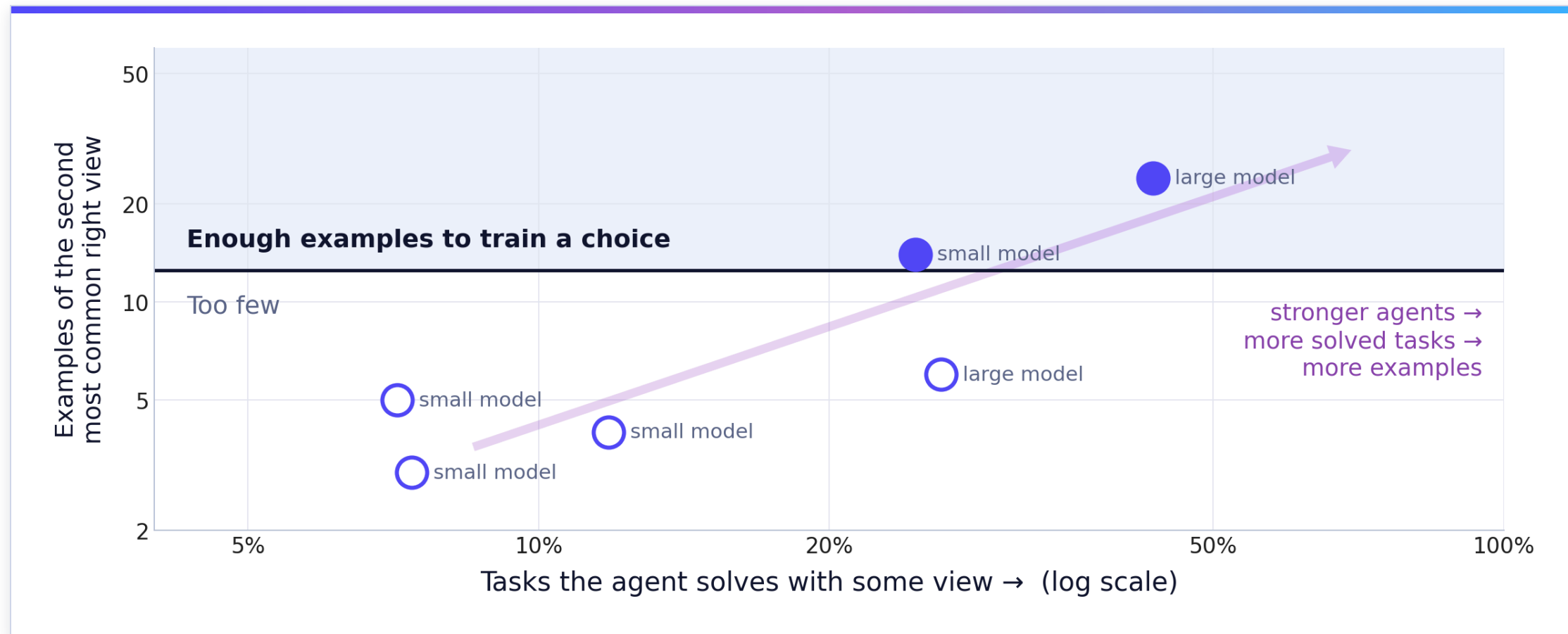
■ No learned router reaches the win corner — even a perfect one rarely does.



Learned routers: 0 of 8 settings

Perfect router: 1 of 8

■ Routing's scaling law: no wins, no examples.



Six VisualWebArena settings · the arrow is a direction, not a fit · below the line, today's agents need at least **2–4×** more tasks

NOT YET

■ Improve the agent first, collect reliable examples, then learn when to look.

WHAT IT TAKES

- a stronger agent
- at least **2–4×** more tasks
- reruns that agree

THREE ASKS

- ✓ type a task at the board
- ✓ which view do you use?
- ✓ lend us a stronger agent

Look, read, or both?
It depends on the task.
Nothing we trained knows it yet.



Come to the board — try your own task, live in all three views.

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REFERENCE – TASK 130, RECORDED STEPPING, 39 S